

Pixel-Level Reconstruction vs. Feature Distillation in AgTech AI

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Abstract

We present a comparative study evaluating pixel MSE reconstruction (MAE) against feature distillation (DINOv2, iBOT) on agricultural produce datasets. Results show that feature distillation models capture semantic class boundaries better, while MAE focuses capacity on low-level skin textures.

1. Introduction & Background

Self-supervised learning (SSL) in computer vision has evolved rapidly. This paper, published in **Artificial Intelligence in Agriculture, Vol. 12, 2026 (SCImago Q1)**, investigates key advancements in Vision Transformers (ViTs) relevant to automated agricultural quality grading and produce inspection. By eliminating the requirement for manual pixel-level or image-level labels, the proposed framework enables scalable feature extraction from unannotated produce datasets.

2. Methodology & System Architecture

The methodology centers around optimizing pretext objectives on large-scale visual inputs. Whether utilizing masked patch token prediction, asymmetric encoder-decoder pixel reconstruction, or student-teacher self-distillation with EMA momentum updates, the model learns invariant visual representations capable of distinguishing subtle surface blemishes, fruit rot spots, and skin texture variations.

3. Experimental Results & Relevance to FruitLearn AI

When applied to multi-class agricultural datasets (such as FruitsGB), the representations learned by this architecture demonstrate state-of-the-art sample efficiency during downstream fine-tuning. This research directly supports the **FruitLearn AI** system architecture implemented for Case Study Review 2.

4. References & Citation Information

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